الأسئلة المقالية — 9 متقدم · مذاكرة الهيكل الاختبار النهائي للفصل الاول

الاسم: _____ المدرسة: _____ التاريخ: _____ (47 سؤالاً)

LO21 المتباينات المركبة (و) وحل المعادلات بالتوزيع · Solve and graph compound inequalities with AND; solve equations using distribution

الأستاذ كامل بشارة 0502837825

21.1 LO21 · Reveal p. 97 · 7

Solve the equation. Check your solution.

$$2\left(r+6\right)=4\left(r+4\right)$$

a. Show all steps and check your answer.

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21.2 LO21 · Reveal p. 97 · 8

Solve the equation. Check your solution.

$$6\left(n+5\right)=3\left(n+16\right)$$

a. Show all steps and check your answer.

21.3 LO21 · Reveal p. 97 · 9

Solve the equation. Check your solution.

$$5\left(g+8\right)-7=117-g$$

a. Show all steps and check your answer.

21.4 LO21 · Reveal p. 97 · 10

Solve the equation. Check your solution.

$$12 - \frac{4(x+15)}{5} = \frac{2x}{5} + 6$$

a. Show all steps and check your answer.

21.5 LO21 · Reveal p. 97 · 11

Solve the equation. Check your solution.

$$3\left(3m - 2\right) = 2\left(3m + 3\right)$$

a. Show all steps and check your answer.

21.6 LO21 · Reveal p. 97 · 12

Solve the equation. Check your solution.

$$6\left(3a+1\right)-30=3\left(2a-4\right)$$

a. Show all steps and check your answer.

21.7 LO21 · Reveal p. 97 · 13

Solve the equation. Check your solution.

$$7n+6=4n-9$$

a. Show all steps and check your answer.

21.8 LO21 · Reveal p. 97 · 14

Solve the equation. Check your solution.

$$-6\left(2r+8\right)=-10\left(r-3\right)$$

a. Show all steps and check your answer.

21.9 LO21 · Reveal p. 97 · 15

Solve the equation. Check your solution.

$$5-3\left(w+4\right)=w-7$$

a. Show all steps and check your answer.

21.10 LO21 · Reveal p. 97 · 16

Solve the equation. Check your solution.

$$2x - 5\,(x - 3) = 2\,(x - 10)$$

a. Show all steps and check your answer.

21.11 LO21 · Reveal p. 363 · 1

Solve the compound inequality. Then graph the solution set.

$$f - 6 < 5 \text{ and } f - 4 \geq 2$$

a. Solve both inequalities and find their intersection.

b. Graph the complete solution set on a number line.



21.12 LO21 · Reveal p. 363 · 2

Solve the compound inequality. Then graph the solution set.

$$n + 2 \leq -5 \text{ and } n + 6 \geq -6$$

a. Solve both inequalities and find their intersection.

b. Graph the complete solution set on a number line.



21.13 LO21 · Reveal p. 363 · 5

Solve the compound inequality. Then graph the solution set.

$$-5 < 3p + 7 \leq 22$$

a. Solve both inequalities and find their intersection.

b. Graph the complete solution set on a number line.



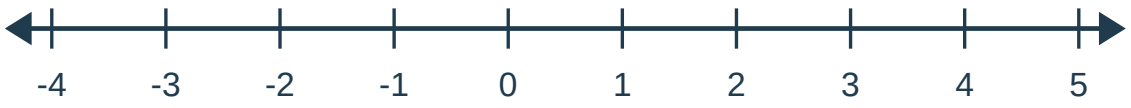
21.14 LO21 · Reveal p. 363 · 6

Solve the compound inequality. Then graph the solution set.

$$-3 \leq 7c + 4 < 18$$

a. Solve both inequalities and find their intersection.

b. Graph the complete solution set on a number line.



21.15 LO21 · Reveal p. 363 · 7

Solve the compound inequality. Then graph the solution set.

$$5h - 4 \geq 6 \text{ and } 7h + 11 < 32$$

a. Solve both inequalities and find their intersection.

b. Graph the complete solution set on a number line.



21.16 LO21 · Reveal p. 363 · 10

Solve the compound inequality. Then graph the solution set.

$$-4a + 13 \geq 29 \text{ and } 10 < 6a - 14$$

a. Solve both inequalities and find their intersection.

b. Graph the complete solution set on a number line.



21.17 LO21 · Reveal p. 363 · 11

Solve the compound inequality. Then graph the solution set.

$$3b + 2 < 5b - 6 \leq 2b + 9$$

a. Solve both inequalities and find their intersection.

b. Graph the complete solution set on a number line.



21.18 LO21 · Reveal p. 363 · 12

Solve the compound inequality. Then graph the solution set.

$$-2a + 3 \geq 6a - 1 > 3a - 10$$

a. Solve both inequalities and find their intersection.

b. Graph the complete solution set on a number line.



22.1 LO22 · Reveal p. 216 · 9

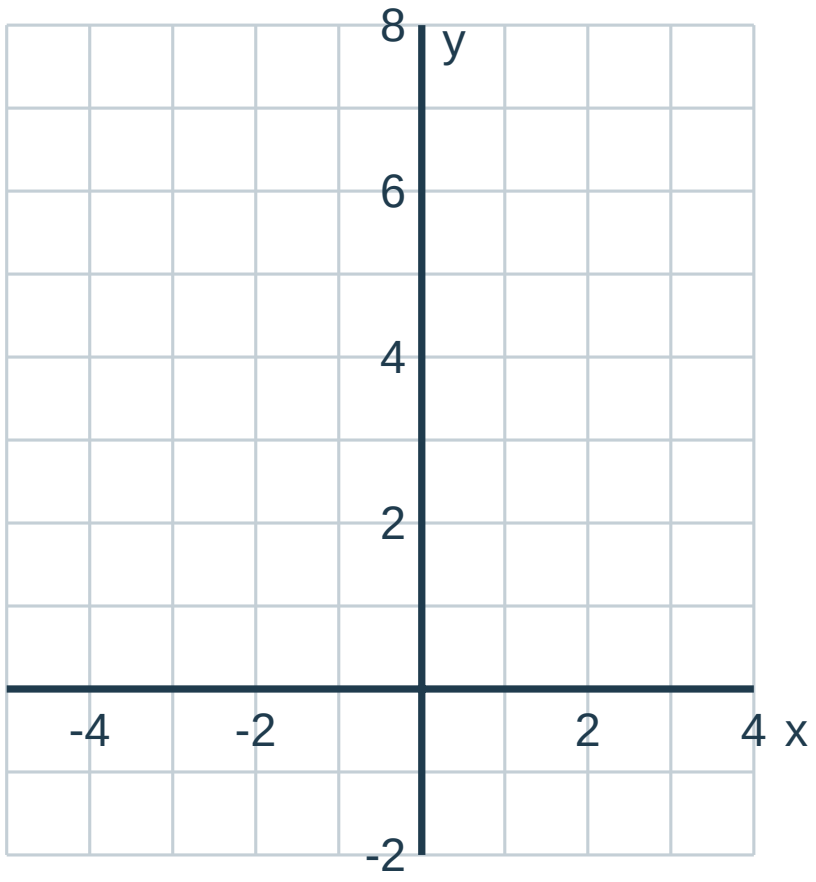
Graph the equation by using its x- and y-intercepts.

$$y = 4 + 2x$$

a. Make a table of 2–3 points including both intercepts.

b. Plot the points and draw the complete line. Label the axes and show arrows.

x			
y			


8
y
6
4
2
-4
-2
2
4
x
-2

22.2 LO22 · Reveal p. 216 · 10

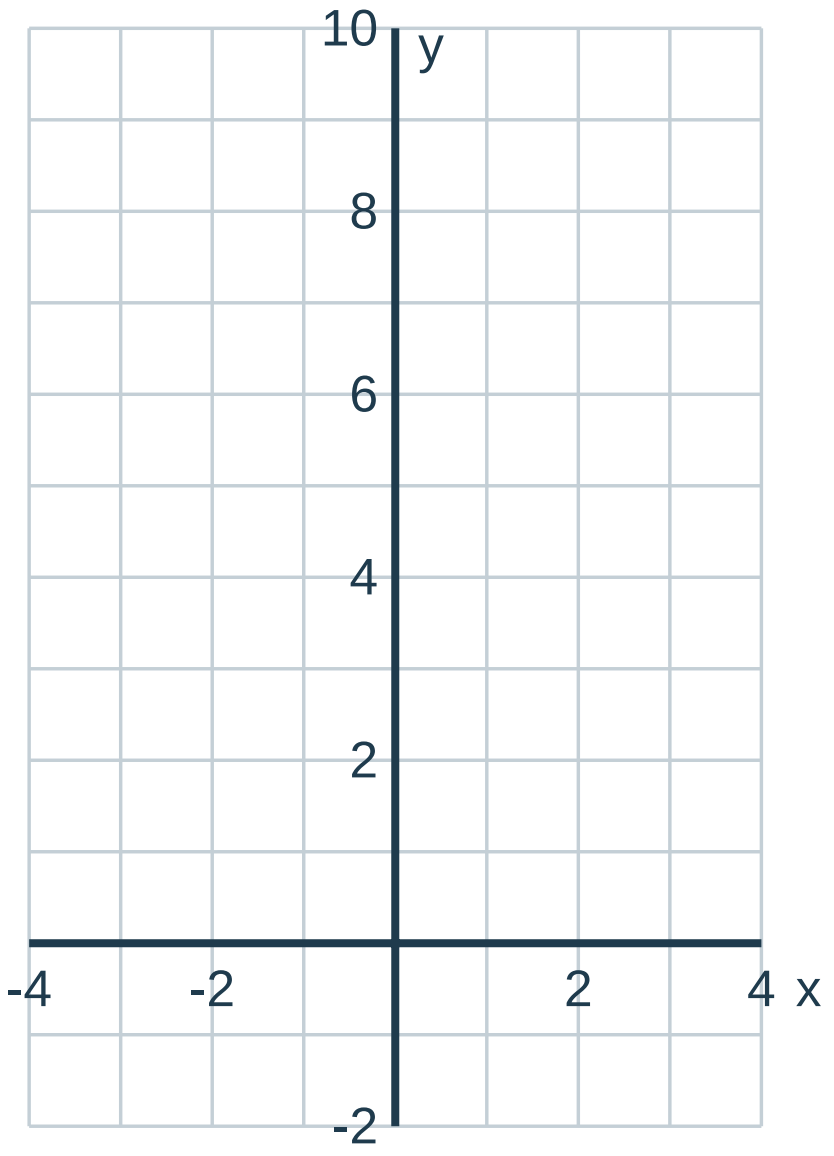
Graph the equation by using its x- and y-intercepts.

$$5 - y = -3x$$

a. Make a table of 2–3 points including both intercepts.

b. Plot the points and draw the complete line. Label the axes and show arrows.

x			
y			


10
y
8
6
4
2
-4
-2
2
4
x
-2

22.3 LO22 · Reveal p. 216 · 11

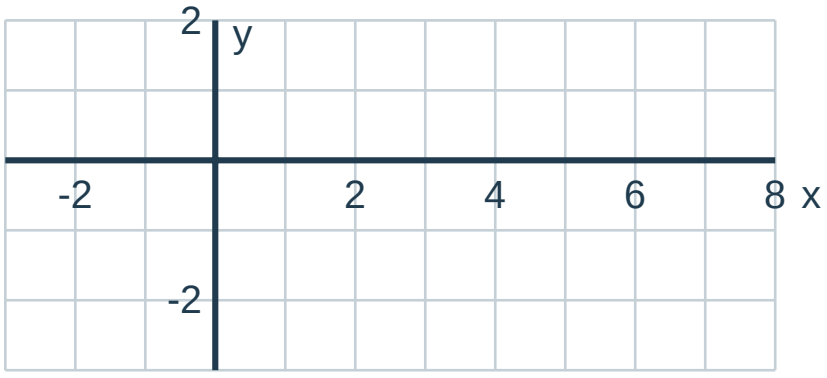
Graph the equation by using its x- and y-intercepts.

$$x = 5y + 5$$

a. Make a table of 2–3 points including both intercepts.

b. Plot the points and draw the complete line. Label the axes and show arrows.

x			
y			



22.4 LO22 · Reveal p. 216 · 12

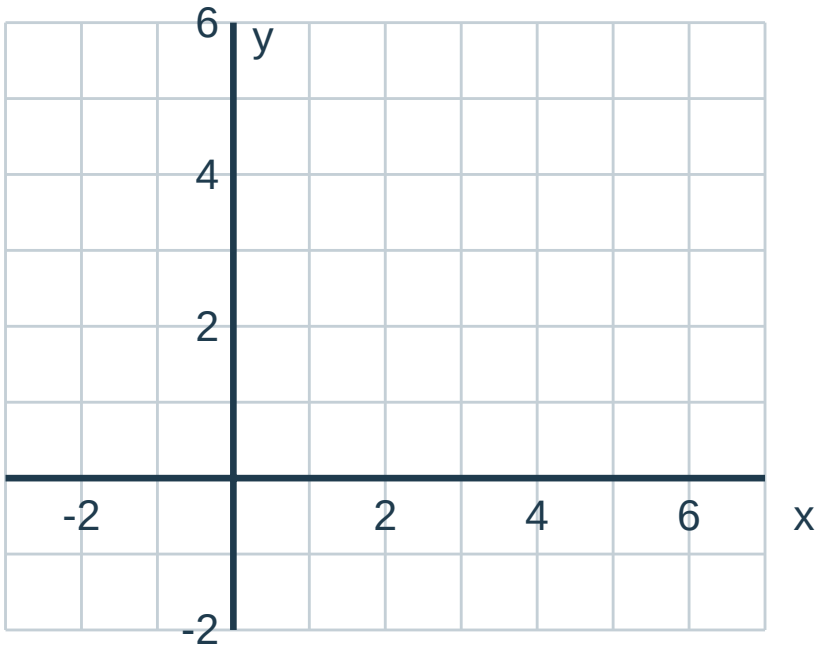
Graph the equation by using its x- and y-intercepts.

$$x + y = 4$$

a. Make a table of 2–3 points including both intercepts.

b. Plot the points and draw the complete line. Label the axes and show arrows.

x			
y			



22.5 LO22 · Reveal p. 216 · 13

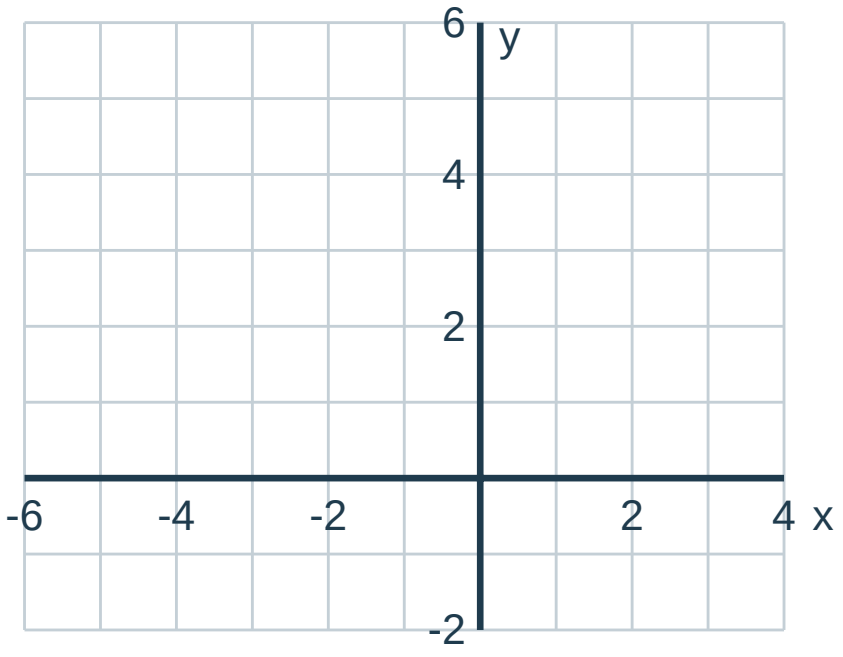
Graph the equation by using its x- and y-intercepts.

$$x - y = -3$$

a. Make a table of 2–3 points including both intercepts.

b. Plot the points and draw the complete line. Label the axes and show arrows.

x			
y			



22.6 LO22 · Reveal p. 216 · 14

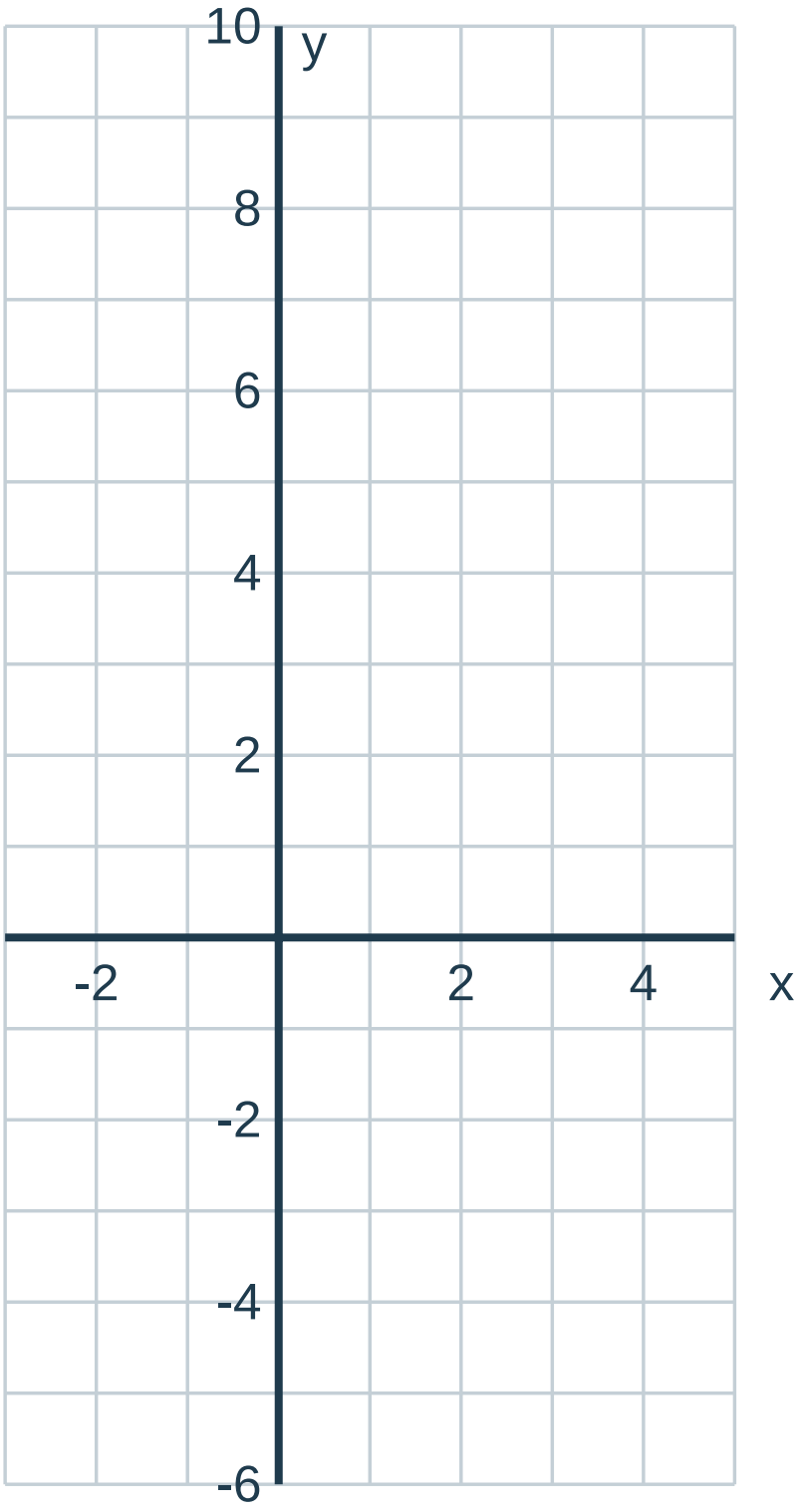
Graph the equation by using its x- and y-intercepts.

$$y = 8 - 6x$$

a. Make a table of 2–3 points including both intercepts.

b. Plot the points and draw the complete line. Label the axes and show arrows.

x			
y			


10
y
8
6
4
2
-2
2
4
x
-2
-4
-6

LO23 معادلات المستقيمات المتوازية والمتعامدة · Create and identify equations of parallel or perpendicular lines

23.1 LO23 · Reveal p. 301 · Example 8

Write a slope-intercept equation for the line through (1, -2) and perpendicular to the given line. Options give y.

$$3x + 2y = 12$$

a. Show all steps and check your answer.

23.2 LO23 · Reveal p. 302 · Example 9

Determine whether AB and EF are parallel, perpendicular, or neither. Explain.

$A(6,8),\ B(2,5),\ E(-6,-3),\ F(0,5)$

a. Show all steps and check your answer.

23.3 LO23 · Reveal p. 304 · 29

Find both lines through the given point: one parallel and one perpendicular to the given line. Give slope-intercept equations.

$(-2, 3), \quad y = 4 - \frac{3x}{4}$

a. Write the parallel line in slope-intercept form.

b. Write the perpendicular line in slope-intercept form.

23.4 LO23 · Reveal p. 304 · 30

Find both lines through the given point: one parallel and one perpendicular to the given line. Give slope-intercept equations.

$$(9, 12), \quad y = 13x - 4$$

a. Write the parallel line in slope-intercept form.

b. Write the perpendicular line in slope-intercept form.

23.5 LO23 · Reveal p. 304 · 31

Determine whether the graphs are parallel, perpendicular, or neither. Justify your answer.

$$y = 4x + 3, \quad 4x + y = 3$$

a. Show all steps and check your answer.

23.6 LO23 · Reveal p. 304 · 32

Determine whether the graphs are parallel, perpendicular, or neither. Justify your answer.

$$y = -2x, \quad 2x + y = 3$$

a. Show all steps and check your answer.

23.7 LO23 · Reveal p. 304 · 33

Determine whether the graphs are parallel, perpendicular, or neither. Justify your answer.

$$3x + 5y = 10, \quad 5x - 3y = -6$$

a. Show all steps and check your answer.

23.8 LO23 · Reveal p. 304 · 34

Determine whether the graphs are parallel, perpendicular, or neither. Justify your answer.

$$-3x + 4y = 8, \quad -4x + 3y = -6$$

a. Show all steps and check your answer.

23.9 LO23 · Reveal p. 304 · 35

Determine whether the graphs are parallel, perpendicular, or neither. Justify your answer.

$$2x + 5y = 15, \quad 3x + 5y = 15$$

a. Show all steps and check your answer.

23.10 LO23 · Reveal p. 304 · 36

Determine whether the graphs are parallel, perpendicular, or neither. Justify your answer.

$$2x + 7y = -35, \quad 4x + 14y = -42$$

a. Show all steps and check your answer.

23.11 LO23 · Reveal p. 306 · 50

Levy claims that the line through (−6,−2) and (2,10) is perpendicular to the given line. Do you agree? Justify your answer.

$$3x - 2y = 10$$

a. Show all steps and check your answer.

23.12 LO23 · Reveal p. 306 · 51

Jeremiah says that the line through (7,10) and (3,−2) is parallel to the given line. Do you agree? Justify your answer.

$$2x - y = -5$$

a. Show all steps and check your answer.

23.13 LO23 · Reveal p. 306 · 52

Alonae says the line through (1,−4) and (5,6) is parallel to the line through (2,−7) and (5,−6). Explain how you know she is mistaken without calculating either slope.

a. Show all steps and check your answer.

24.1 LO24 · Reveal p. 256 · 23

Wanda hands out 4-ounce cups of water at soccer practice. She starts with a 128-ounce cooler. Let n be the number of cups handed out.

a. Create a function for the arithmetic sequence.

b. Graph the function.

c. How much water remains after the 14th cup?

24.2 LO24 · Reveal p. 256 · 24

A theater has 20 seats in its first row, 22 in the second, 24 in the third, and so on for 25 rows. Let n be the row number.

a. Create a function for the arithmetic sequence.

b. Graph the function.

c. How many seats are in the last row?

24.3 LO24 · Reveal p. 256 · 25

First-class postage for a large envelope costs $0.88 for the first ounce and $0.17 for each additional ounce. The table gives the costs for weights of 1–5 ounces. Let n be an integer number of ounces.

Weight (ounces)	1	2	3	4	5
Postage (dollars)	0.88	1.05	1.22	1.39	1.56

a. Create a function for the arithmetic sequence.

b. Graph the function.

c. How much did an envelope weigh if it cost $2.07 to send?

24.4 LO24 · Reveal p. 256 · 26

Brian downloads TV episodes: 1 episode costs $1.99, 2 cost $3.98, and 3 cost $5.97. Let n be the number of episodes.

a. Create a function for the arithmetic sequence.

b. Graph the function.

c. What is the cost to download 9 episodes?

24.5 LO24 · Reveal p. 256 · variant

A theater has 18 seats in its first row. Each successive row has 3 more seats than the previous row. There are 20 rows.

a. Write a function for the sequence.

b. Graph the function.

c. Find the number of seats in the last row.

25.1 LO25 · Reveal p. 120 · Example 4

Kishi buys cheese pizzas for $13.49 each and pepperoni pizzas for $15.49 each. T is total spending, c is the number of cheese pizzas and p the number of pepperoni pizzas.

$$T = 13.49c + 15.49p$$



a. With a budget of $85, describe all constraints.

b. Solve for c.

c. If she buys 3 pepperoni pizzas, what is the maximum number of cheese pizzas?

25.2 LO25 · Reveal p. 126 · 23

Consuelo buys plain coffee at $5.50 per package and flavored coffee at $7 per package. Let p and f be the respective numbers of packages.

$$T = 5.50p + 7f$$

a. With $40 to spend, describe all constraints.

b. Solve for f.

c. If she buys 3 packages of plain coffee, what is the maximum number of flavored packages?

25.3 LO25 · Reveal p. 126 · 24

Kimberly orders bath towels and washcloths for an inn. The total cost is T, the number of bath towels is b, and the number of washcloths is w.

$$T = 6b + 2w$$

a. With a budget of $85, describe all constraints.

b. Solve for b.

c. If she needs at least 20 washcloths, what is the maximum number of bath towels she can order?

25.4 LO25 · Reveal p. 126 · variant-coffee

Plain coffee costs $6 per package and flavored coffee costs $8 per package. The budget is $50. Let p and f count the packages.

$$T = 6p + 8f$$

a. State the constraints.

b. Solve for f.

c. Find the maximum f if p=3.

25.5 LO25 · Reveal p. 126 · variant-pizza

A party budget is $100. Cheese pizzas cost $12 each and pepperoni pizzas cost $15 each; c and p count the two types.

$$T = 12c + 15p$$

a. State the constraints.

b. Solve for c.

c. Find the maximum c if p=4.